GAZE LINK MANUAL

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What is Gaze Link?

Gaze Link is a multi-language low-cost eye-gesture communication system intended for People with Amyotrophic lateral sclerosis (PALS) who develop severe motor and verbal disabilities. The application includes a text-entry keyboard that allows PALS to type grammatical sentences independently and other functionalities for calibration, settings adjustments, and quick phrase entry. The system uses Large Language Models (LLMs) on the cloud for context-aware sentence generation and word prediction tools for a higher text-entry rate. The application is currently in alpha testing phase.

Youtube Channel: https://www.youtube.com/@GazeLink-ru7em/featured
Google Play Store Download:
https://play.google.com/store/apps/details?id=com.demo.opencv&hl=en_US

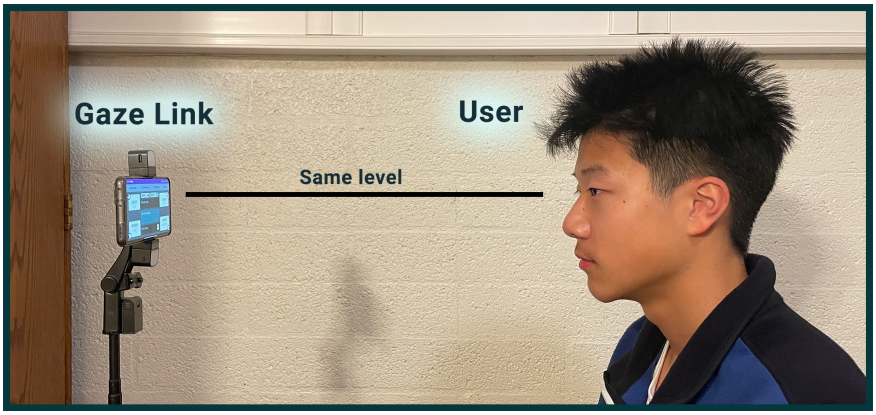
Functionalities:

1. A text-entry mode for typing words and sentences with 6 eye gestures
2. A quick chat mode for speaking pre-stored phrases quickly
3. A calibration mode to set up the user's eye gesture recognition
4. A settings mode to personalize settings
5. A data mode to see some data with the eye gestures



Step 1: Setup

A tripod or stand is recommended to stabilize the device. An assistant can also hold the phone in front of the user. The user should be seated and facing the front camera. The distance between the user's eyes and the display depend on the user as long as the app can detect the eyes accurately.



Step 2: Calibration

After opening the app, navigate to **calibration mode** by clicking on the “CALIBRATE” button on the topbar. For MND patients, this process requires an **assistant** to complete.



After entering calibration mode, ensure that the user is facing the phone’s screen, and the system can detect the eyes.

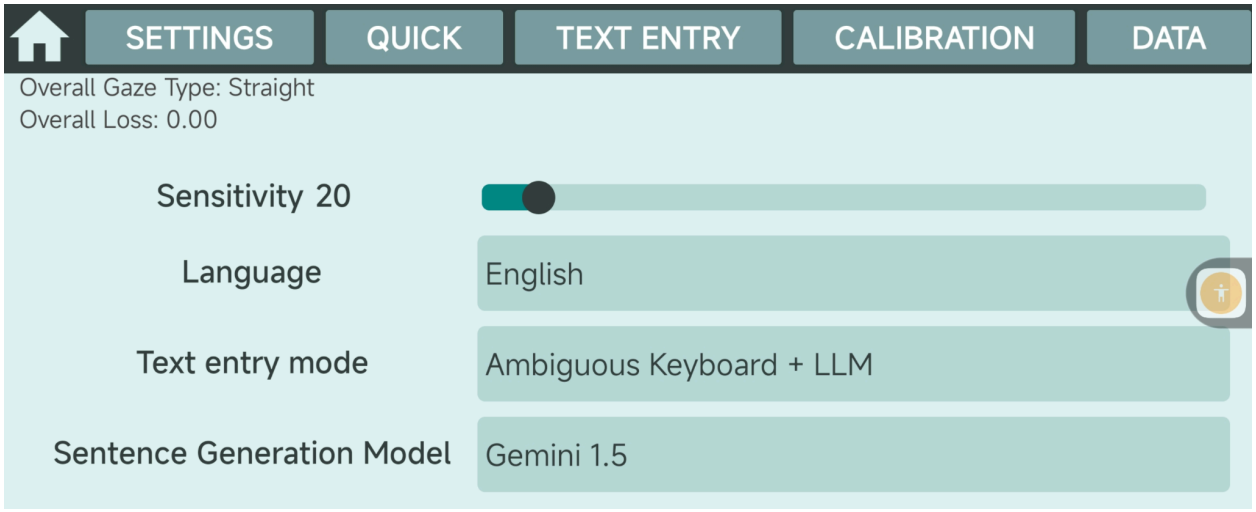
1. Click continue to begin, and the system will display a small frame that contains your real-time eye frame.
2. Follow the instruction on the top left of the screen or a voice instruction
3. When the user looks in the correct direction, the assistant will click the “continue” button to capture the patient’s eye frame.
4. Repeat the process until all 6 frames are collected

Calibration is required for the first time opening the app and recommended when the user or lighting condition changes.

Calibration video demonstration:
https://drive.google.com/file/d/14jswX6tfBh-v1QTUEAx59v5hDLXppByH/view?usp=sharing

Step 3: Configurations

After calibrating the patient's eyes, navigate to the settings mode by clicking on the "SETTINGS MODE" button in the topbar.



Display Information:

Display Name	**Function**
Gaze Input Log	Shows the previous gestures inputted by the user
Language	Choose between English, Spanish, and Chinese.
Sentence Generation Model	Choose between different LLM models for sentence generation.
Overall Gaze Type	Shows the system's gesture recognition of the *current frame*. The recognition for the current frame does not mean that an input is

	performed. The system uses the overall gaze type of multiple frames to determine whether the user's gaze is intentional or not.
Overall Loss	The mathematical difference between the calibration frame and the user's real-time eye frame. A higher loss (> 0.03) means that the two images are less similar, while a lower loss (< 0.02) means the images are more similar.
Gaze Sensitivity	The threshold for input detection. A higher gaze sensitivity (> 0.3) means that gaze gestures will be detected more easily, but false detections will also be more common. A lower gaze sensitivity (< 0.2) means that gaze gestures will be detected less easily, but detections will be more accurate.
Current Text-entry Mode	**Mode 1: letter-by-letter keyboard.** In this mode, you can only type letters one by one with no word or sentence prediction. Use this mode to type complicated words that cannot be typed in previous modes. **Mode 2: ambiguous keyboard only (baseline).** In this mode, you will type words with the ambiguous keyboard but without Large Language Models (LLMs) for sentence prediction. **Mode 3: ambiguous keyboard + LLM.** This mode implements LLMs in the system for word bank extension and sentence retrieval. Requires the internet to function.

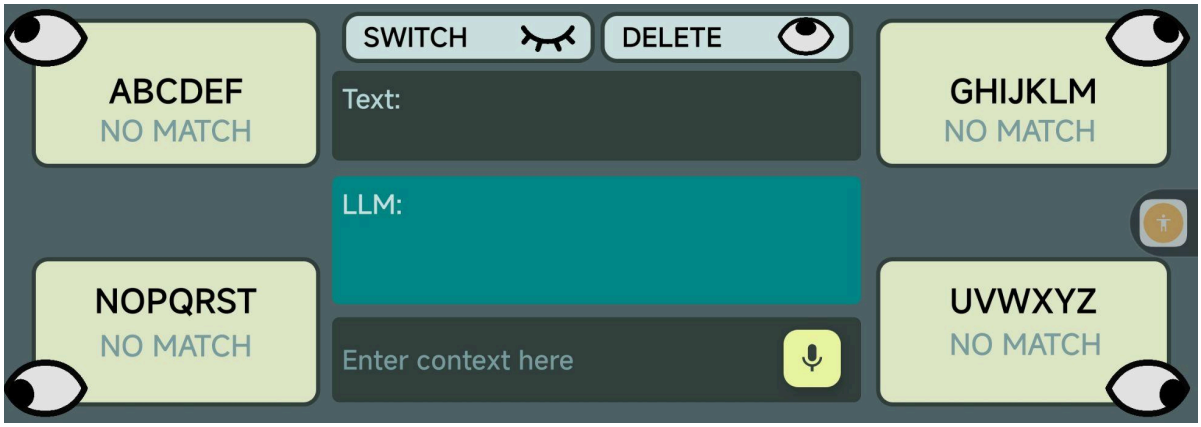
After entering settings mode and completing calibration, the system should begin to recognize the gaze of the patient. Instruct the patient to look in all directions (up, left up, right up, left, right, closed) and check the **Gaze Input Log** in the bottom right corner. If the gaze gesture is recognized (you will hear a *beep* sound), the detection is working well.

However, if the gaze gesture is not registered, check the **Overall Gaze Type** and **Overall Loss** on the top left corner.

1. If the overall gaze type is incorrect, then the calibration frames are too similar. The calibration should be redone with more varied eye movements.
2. If the overall loss is high (> 0.03), turn up the gaze sensitivity until the gaze gesture is registered.

Step 4: Text-entry system

Navigate to the text-entry system by clicking on the "TEXT-ENTRY MODE" button on the topbar.



Here is an overview of each gaze gesture's function:

	Letter Mode	**Word Mode**
Left-up	Type "ABCDEF"	Type word 1
Right-up	Type "GHIJKLM"	Type word 2
Left-down	Type "NOPQRST"	Type word 3
Right-down	Type "UVWXYZ"	Go to next page
Closed	Switch to word mode	Switch to letter mode
Up	Delete (letters or words)	Finish (for text-to-speech or social media upload)

Before text-entry, the assistant or conversation partner should record the dialogue context into the system by manually entering text or clicking the voice recording button.

Gaze Link uses an ambiguous keyboard system to improve text-entry rate. Instead of specifically typing a letter, the user only has to perform the eye gesture that contains the letter. For example, to type “C”, the user should look left-up to type “ABCDEF”. To type “S”, the user should look left-down to type “NOPQRST”.

After typing all the letters of a word, the system will automatically generate suitable word options. The user should perform a “switch” action with the closed eye gesture. Then, the user can select the word from the 4 options on each corner. After selecting the word, the system will generate a sentence based on user input if LLM is enabled. Speak the sentence with the “speak” button.

Text-entry mode also supports touchscreen. If the keyboard malfunctioned and caused unnecessary inputs, the assistant can delete the extra inputs as well.

Text-entry system tutorial:
https://www.youtube.com/watch?v=-Kv0H2DQ-Y8

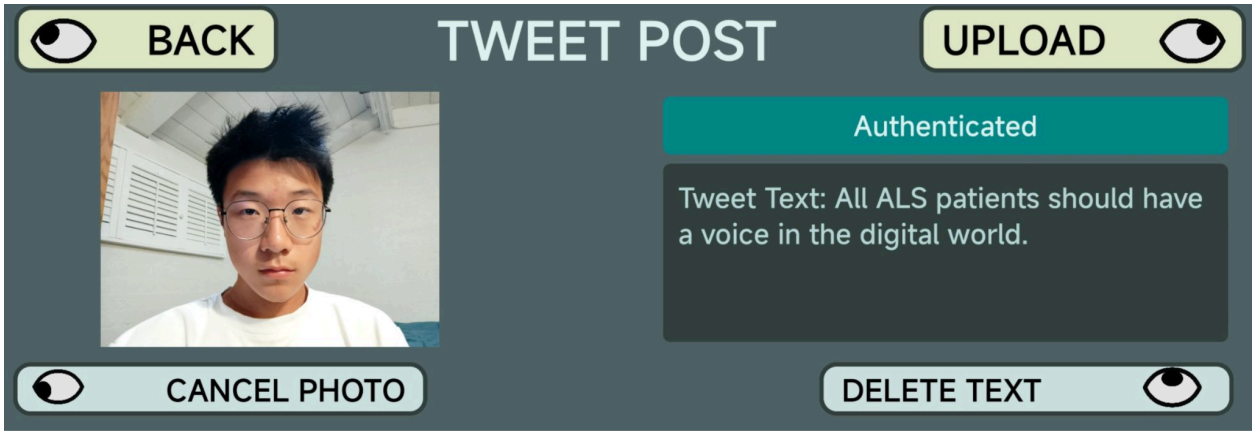
Step 5: Social Media Upload

After finishing the sentence, the user can choose between uttering the text with the phone’s speaker or uploading it to social media (Twitter).



Speak text (look left down), Upload to social media (look right down)

After looking right down, the app will go in the social media interface.



If not authenticated, press the green button beneath upload to authenticate with Twitter. After authenticating, you can upload the text directly to your account. You can also send a picture by taking a selfie with the phone's front camera.